# CPSC 4420/6420: ARTIFICIAL INTELLIGENCE ASSIGNMENT 2

DUE: Sunday, Oct 18, 2026 @ 11:59 PM NAME: _____

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For this assignment, select **one** of the following **Gymnasium environments** and develop a Python program that uses **Q-learning** to learn a policy. Investigate how the algorithm behaves under different hyperparameter settings. The environment choices are:

1. **FrozenLake**: Navigate to the goal while avoiding holes. Use the $8 \times 8$ map.
2. **Taxi**: Pick up a passenger and deliver them to the correct destination.
3. **CliffWalking**: Reach the goal without falling off the cliff.
4. **Blackjack**: Learn when to hit or stick to maximize expected reward.
5. **MountainCar**: Build momentum to reach the hilltop. Discretize the continuous observations for tabular Q-learning.

(A) [10 pts] **Environment description.** Describe the environment's goal, state/observation space, action space, transition dynamics, reward structure, and termination conditions. Explain any state discretization or environment settings you use.

(B) [25 pts] **Implement Q-learning.** Write your own tabular Q-learning agent, including Q-value initialization, an $\epsilon$-greedy action-selection policy, a training loop, terminal-state handling, and the update

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[ r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right].$$

Make sure the learning rate $\alpha$, discount factor $\gamma$, and exploration parameters easy to change.

(C) [10 pts] **Baseline training.** Choose and report a baseline configuration ($\alpha$, $\gamma$, $\epsilon$, number of episodes, and any exploration schedule). Train the agent and include a plot of reward versus training episode. Explain your initial results.

(D) [15 pts] **Learning-rate experiment.** Test at least **three** values of $\alpha$ (e.g., $0.1,\ 0.5,\ 0.9$) while keeping the other settings fixed. Plot and compare learning speed, stability, and final performance. Explain your observations.

(E) [15 pts] **Discount-factor experiment.** Test at least **three** values of $\gamma$ (e.g., $0.50,\ 0.90,\ 0.99$), keeping the other settings fixed. Present a comparison plot and discuss the effect of valuing immediate versus future rewards.

(F) [10 pts] **One additional hyperparameter.** Vary at least **three** settings of one other parameter (e.g., $\epsilon$, $\epsilon$-decay, training episodes, or state discretization). Include a comparison plot and explain the effect on learning.

(G) [15 pts] **Evaluate your best agent.** Evaluate the final trained policy over at least **30 episodes**, with exploration disabled. Report mean reward and appropriate additional metrics (e.g., success rate or steps to completion). Compare the results to your baseline. Explain how you selected the best configuration.

(H) **[10 pts – Grad only; extra credit for undergrad] Stochastic environment.** Modify the environment to introduce a controlled source of stochasticity. For example, with probability $p$, replace the agent's intended action with a *different* valid action. Test at least three noise levels, such as $p = 0,\ 0.10,\ 0.25$. Train and evaluate under each noise level, and plot how the uncertainty affects performance. Also test whether your original ($p = 0$) trained policy remains effective under added noise. Discuss the effect on learning speed, final reward, and policy robustness. If your chosen environment is already stochastic, add a separate, controllable source of uncertainty.

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**Submission:** Upload a **PDF report** to Canvas describing the environment, method, experimental settings, figures, results, and conclusions. Include a link to a **public GitHub repository** with your code, `README.md` (setup and run instructions), results/plots, and a rendered **MP4 video** of the trained agent. Do **not** upload individual code files to Canvas. Make sure your repository link is accessible.